Department of Mathematics, IIT Madras

MA2101 Linear Algebra via Matrices

Assignment 3

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1. Let U be a subspace of V and let V be a subspace of W, where W is a subspace of $\mathbb{F}^n$. Is U a subspace of W?
2. Let $u, v_1, v_2, \ldots, v_n$ be $n+1$ distinct vectors in $\mathbb{F}^n$, $S_1 = \{v_1, v_2, \ldots, v_n\}$, and let $S_2 = \{u, v_1, v_2, \ldots, v_n\}$. Prove that $\text{span}(S_1) = \text{span}(S_2)$ iff $u \in \text{span}(S_1)$.
3. Let A and B be subsets of $\mathbb{F}^n$. Prove or disprove the following:
   (a) A is a subspace of $\mathbb{F}^n$ if and only if $\text{span}(A) = A$.
   (b) If $A \subseteq B$, then $\text{span}(A) \subseteq \text{span}(B)$.
   (c) $\text{span}(A \cup B) = \{u + v : u \in \text{span}(A),\ v \in \text{span}(B)\}$.
   (d) $\text{span}(A \cap B) \subseteq \text{span}(A) \cap \text{span}(B)$.
   (e) $\text{span}(A) \cap \text{span}(B) \subseteq \text{span}(A \cap B)$.
4. Let $S \subseteq U \subseteq \mathbb{F}^n$, where U is a subspace of $\mathbb{F}^n$. Suppose for any subspace V of $\mathbb{F}^n$, if $S \subseteq V$ then $U \subseteq V$. Prove that $U = \text{span}(S)$.
5. Let A and B be subsets of $\mathbb{F}^n$. Prove or disprove: $A \cup B$ is linearly independent iff $\text{span}(A) \cap \text{span}(B) = \{0\}$.
6. Let V be a subspace of $\mathbb{F}^n$ and let $B \subseteq V$. Prove the following:
   (a) B is a basis of V iff B is linearly independent and each proper superset of B is linearly dependent.
   (b) B is a basis of V iff B spans V and no proper subset of B spans V.
7. Let $\{x, y, z\}$ be a basis for a vector space V. Is $\{x + y, y + z, z + x\}$ also a basis for V?
8. Find a basis for the subspace $\{(a, b, c) \in \mathbb{R}^3 : a + b - 5c = 0\}$ of $\mathbb{R}^3$.
9. Find bases and dimensions of the following subspaces of $\mathbb{R}^5$:
   (a) $\{(a, b, c, d, e) \in \mathbb{R}^5 : a - c - d = 0\}$.
   (b) $\{(a, b, c, d, e) \in \mathbb{R}^5 : b = c = d,\ a + e = 0\}$.
   (c) $\text{span}\{(1, -1, 0, 2, 1),\ (2, 1, -2, 0, 0),\ (0, -3, 2, 4, 2),$
   $(3, 3, -4, -2, -1),\ (5, 7, -3, -2, 0)\}$.
10. Extend the set $\{(1, 0, 1, 0),\ (1, 0, -1, 0)\}$ to a basis of $\mathbb{R}^4$.
11. Let $u_1, v, w, x_1, x_2, x_3, x_4 \in \mathbb{R}^5$ satisfy $x_1 = u + 2v + 3w$, $x_2 = 2u - 3v + 4w$, $x_3 = -3u + 4v - 5w$ and $x_4 = -u + 6v + w$. Is $\{x_1, x_2, x_3, x_4\}$ linearly dependent?
12. Prove that the only proper subspaces of $\mathbb{R}^2$ are the straight lines passing through the origin.
13. Describe all subspaces of $\mathbb{R}^3$.
14. Let $A \in \mathbb{F}^{n \times n}$. Let $\{v_1, \ldots, v_n\}$ be any basis of $\mathbb{F}^{n \times 1}$. Prove that A is invertible iff $Ax = v_i$ has at least one solution for each $i = 1, \ldots, n$.
15. Let $A = \begin{bmatrix} 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ 1 & 1 & 1 & 2 & 1 \\ 5 & 8 & 5 & 7 & 5 \end{bmatrix}$. Write $r = \text{rank}(A)$ and $k = \text{null}(A)$.
   (a) Find bases for $R(A)$ and $N(A)$, and then determine r and k.
   (b) Express suitable $4 - r$ rows of A as linear combinations of other r rows.
   (c) Express suitable $5 - r$ columns of A as linear combinations of other r columns.
16. Construct bases for (i) the row space, (ii) the column space, and the null space for each of the following matrices:
   (a) $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 7 \\ 2 & 4 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & 1 & -3 \\ 1 & 2 & 8 \\ 3 & -1 & 4 \\ 4 & -2 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 2 \\ 1 & 3 & -2 & 1 \end{bmatrix}$

17. Let $A \in \mathbb{F}^{n\times n}$. Consider A as the linear transformation $A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{n\times 1}$ given by $A(x) = Ax$ for each $x \in \mathbb{F}^{n\times 1}$. Prove the following:

(a) A is one-one iff $N(A) = \{0\}$.
(b) A is one-one and onto iff A maps any basis onto a basis of $\mathbb{F}^{n\times 1}$.
(c) A is one-one iff A is onto.

18. Let A and B be matrices having the same RREF. Then which of the following may be concluded? Explain.
(a) $R(A) = R(B)$ (b) $N(A) = N(B)$
(c) The row space of A is equal to the row space of B.

19. For $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times k}$, prove: $\text{rank}(AB) \le \min\{\text{rank}(A), \text{rank}(B)\}$.

20. In the following a basis B for $\mathbb{F}^{3\times 1}$ and a vector $u \in \mathbb{F}^{3\times 1}$ are given. Compute the coordinate vector $[u]_B$.

(a) $B = \{[0,0,1]^t,\ [1,0,0]^t,\ [0,1,0]^t\},\ u = [1,2,3]^t$
(b) $B = \{[0,1,1]^t,\ [1,0,1]^t,\ [1,1,0]^t\},\ u = [2,1,3]^t$
(c) $B = \{[1,1,1]^t,\ [1,2,1]^t,\ [1,2,3]^t\},\ u = [3,2,1]^t$

21. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$. Consider the bases $B = \{[1,0,1]^t,\ [0,1,1]^t,\ [1,1,0]^t$ and $C = \{[1,1,1]^t,\ [1,2,1]^t, [1,1,0]^t\}$ for $\mathbb{F}^{3\times 1}$. Compute $[A]_{C,B}$ and $[A]_{B,C}$ in the following ways:

(a) by expressing A-images of the basis vectors of B in terms of those of C and vice versa.
(b) by computing the inverses of two matrices constructed from the basis vectors.
(c) by using the RREF conversion as mentioned in the text.

22. Let $A, B \in \mathbb{F}^{m\times n}$, C be a basis of $\mathbb{F}^{n\times 1}$, D be a basis for $\mathbb{F}^{m\times 1}$, and let $\alpha \in \mathbb{F}$. Show that $[\alpha A + B]_{D,C} = \alpha[A]_{D,C} + [B]_{D,C}$.

23. Construct a matrix $A \in \mathbb{R}^{2\times 2}$, a vector $v \in \mathbb{R}^{2\times 1}$, and a basis $B = \{u_1, u_2\}$ for $\mathbb{R}^{2\times 1}$ satisfying $[Av]_B \neq A[v]_B$.

24. Let $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 1 & 3 \\ -1 & -1 & -2 \end{bmatrix}$, $V = \{[a,b,c]^t : a+2b+3c = 0,\ a,b,c \in \mathbb{R}\}$, $B = \{[0,3,-2]^t,\ [2,-1,0]^t\}$, and let $C = \{[1,1,-1]^t,\ [3,0,-1]^t\}$.

(a) Show that V is a 2-dimensional subspace of $\mathbb{R}^{3\times 1}$.
(b) Show that $A : V \to V$ defined by $A(v) = Av$ for $v \in V$ is a well defined function.
(c) Show that B and C are bases for V.
(d) Extend the basis B of V to a basis O for $\mathbb{F}^{3\times 1}$.
(e) Extend the basis C of V to a basis N for $\mathbb{F}^{3\times 1}$.
(f) Find the change of basis matrix $I_{N,O}$.
(g) Verify that $[v]_N = I_{N,O}\,[v]_O$ for $v = [-4,-1,2]^t$.
(h) Find the matrix $[A]_{N,O}$.
(i) Verify that $[Av]_N = [A]_{N,O}\,[v]_O$ for $v = [-4,-1,2]^t$.

25. In each of the following cases, show that T is a linear transformation. Find the matrix A so that $T(x) = Ax$. Determine $\text{rank}(A)$. And then construct the full rank factorization of A.

(a) $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{2\times 1},\ T([a,b,c]^t) = [c, b+a]^t$
(b) Let $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{3\times 1},\ T([a,\ b,\ c]^t) = [a+b,\ 2a-b-c,\ a+b+c]^t$

26. Which matrices are equivalent to, and which are similar to
(a) the zero matrix. (b) the identity matrix?

27. Consider the bases $O = \{u_1, u_2\}$ and $N = \{v_1, v_2\}$ for $\mathbb{F}^{2\times 1}$, where $u_1 = [1,1]^t, u_2 = [-1,1]^t, v_1 = [2,1]^t$, and $v_2 = [1,0]^t$. Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$.

(a) Compute $Q = [A]_{O,O}$ and $R = [A]_{N,N}$.
(b) Find the change of basis matrix $P = I_{N,O}$.
(c) Compute $S = PQP^{-1}$.
(d) Is it true that $R = S$? Why?
(e) If $S = [s_{ij}]$, verify that $Av_1 = s_{11}v_1 + s_{21}v_2,\ Av_2 = s_{12}v_1 + s_{22}v_2$.

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2. Let $u, v_1, v_2, \ldots, v_n$ be $n+1$ distinct vectors in $\mathbb{F}^n$, $S_1 = \{v_1, v_2, \ldots, v_n\}$, and let $S_2 = \{u, v_1, v_2, \ldots, v_n\}$. Prove that $\text{span}(S_1) = \text{span}(S_2)$ iff $u \in \text{span}(S_1)$.

Since S_2 has all elements of S_1 including u;

$\text{span}(S_1) \subseteq \text{span}(S_2)$

Given $\text{span}(S_1) = \text{span}(S_2) \Rightarrow \text{span}(S_2) \subseteq \text{span}(S_1)$

$\Rightarrow$ every vector in $\text{span}(S_2)$ must be in $\text{span}(S_1) \Rightarrow u \in \text{span}(S_1)$

If $u \in \text{span}(S_1) \Rightarrow u = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$

any vector in $\text{span}(S_2)$ can be written as

$w = \beta u + \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_n v_n$

$= (\beta\alpha_1 + \beta_1) v_1 + (\beta\alpha_2 + \beta_2) v_2 + \cdots + (\beta\alpha_n + \beta_n) v_n$

= linear combination of $v_1, v_2, \ldots v_n$

$\Rightarrow w \in \text{span}(S_1) \Rightarrow \text{span}(S_2) \subseteq \text{span}(S_1)$

But since $\text{span}(S_1) \subseteq \text{span}(S_2) \Rightarrow \text{span}(S_1) = \text{span}(S_2)$

(a) A is a subspace of $\mathbb{F}^n$ if and only if $\text{span}(A) = A$.
(b) If $A \subseteq B$, then $\text{span}(A) \subseteq \text{span}(B)$.
(c) $\text{span}(A \cup B) = \{u + v : u \in \text{span}(A),\ v \in \text{span}(B)\}$.
(d) $\text{span}(A \cap B) \subseteq \text{span}(A) \cap \text{span}(B)$.
(e) $\text{span}(A) \cap \text{span}(B) \subseteq \text{span}(A \cap B)$.

(a) If A is a subspace of F^n, A contains all linear combinations of its elements

$\Rightarrow \text{span}(A) \subseteq A$. But span is the smallest

Subspace containing $A \Rightarrow \text{span}(A) = A$

If $\text{span}(A) = A \Rightarrow A$ is closed under linear combinations $\Rightarrow A$ is subspace of B.

(b) $A \subseteq B \Rightarrow$ every LCOA is in LCOB $\Rightarrow \text{span}(A) \subseteq \text{span}(B)$

(d) $A \cap B$ is in both A and B

$\Rightarrow$ LCO$(A \cap B)$ is in LCOA and LCOB

$\Rightarrow$ span$(A \cap B)$ is in both span(A) and span(B)

(e) False

4. Let $S \subseteq U \subseteq \mathbb{F}^n$, where U is a subspace of $\mathbb{F}^n$. Suppose for any subspace V of $\mathbb{F}^n$, if $S \subseteq V$ then $U \subseteq V$. Prove that $U = \text{span}(S)$.

$S \subseteq U$ U contains all the linear combinations of elements of S

$\Rightarrow \text{span}(S) \subseteq U$

Let $V = \text{span}(S) \Rightarrow S \subseteq \text{span}(S)$ ✓

$\Rightarrow U \subseteq \text{span}(S)$

$\Rightarrow U = \text{span}(S)$

5. Let A and B be subsets of $\mathbb{F}^n$. Prove or disprove: $A \cup B$ is linearly independent iff $\text{span}(A) \cap \text{span}(B) = \{0\}$.

Let $v \in \text{span}(A) \cap \text{span}(B)$

$\Rightarrow v = \sum_i \alpha_i a_j = \sum_j \beta_j b_j \quad a_i \in A, b_j \in B$

$\Rightarrow \sum_i \alpha_i a_i - \sum_j \beta_j b_j = 0$

But $A \cup B$ is linearly independent $\Rightarrow \alpha_i = \beta_j = 0$

$\Rightarrow 0 = 0$

7. Let $\{x, y, z\}$ be a basis for a vector space V. Is $\{x + y, y + z, z + x\}$ also a basis for V?

Yes, they're linearly independent and spans over V

8. Find a basis for the subspace $\{(a, b, c) \in \mathbb{R}^3 : a + b - 5c = 0\}$ of $\mathbb{R}^3$.

$a = -b + 5c \Rightarrow (-b+5c, b, c) = b(-1,1,0) + c(5,0,1)$

9. Find bases and dimensions of the following subspaces of $\mathbb{R}^5$:

   (a) $\{(a, b, c, d, e) \in \mathbb{R}^5 : a - c - d = 0\}$.
   (b) $\{(a, b, c, d, e) \in \mathbb{R}^5 : b = c = d,\ a + e = 0\}$.
   (c) $\text{span}\{(1, -1, 0, 2, 1),\ (2, 1, -2, 0, 0),\ (0, 3, 2, 4, 2),\ (3, 3, -4, -2, -1),\ (5, 7, -3, -2, 0)\}$.

→ write as column

(4)

(a) $(c+d, b, c, d, e) = b(0,1,0,0,0) + c(1,0,1,0,0) + d(1,0,0,1,0) + e(0,0,0,0,1)$

(2)

(b) $(a, b, b, b, -a) = a(1,0,0,0,-1) + b(0,1,1,1,0)$

(c)

$$\begin{matrix} 1 & -1 & 0 & 2 & 1 \\ 2 & 1 & -2 & 0 & 0 \\ 0 & -3 & 2 & 4 & 2 \\ 3 & 3 & -4 & -2 & -1 \\ 5 & 7 & -3 & -2 & 0 \end{matrix} \xrightarrow[E_3[4,1]\ E_{-5}[5,1]]{E_{-2}[2,1]} \begin{matrix} 1 & -1 & 0 & 2 & 1 \\ 0 & 3 & -2 & -4 & -2 \\ 0 & -3 & 2 & 4 & 2 \\ 0 & 6 & -4 & -8 & -4 \\ 0 & 12 & -3 & -12 & -5 \end{matrix}$$

$$\downarrow \quad E_{1/3}[1,2]\ \ E_{-2}[4,2]\ \ E_1[3,2]\ \ E_{-4}[5,2]$$

$$\begin{matrix} 1 & 0 & -2/3 & 2/3 & 1/3 \\ 0 & 1 & -2/3 & -4/3 & -2/3 \\ 0 & 0 & 1 & 4/5 & 3/5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{matrix} \xleftarrow[E[3,5]\ E_{1/5}[3]]{E_{1/3}[2]} \begin{matrix} 1 & 0 & -2/3 & 2/3 & 1/3 \\ 0 & 3 & -2 & -4 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 4 & 3 \end{matrix}$$

$E_{3,5}[1,3]$
$E_{3,5}[2,3]$

$$\begin{matrix} 1 & 0 & 0 & 28/15 & 37/30 \\ 0 & 1 & 0 & -2/15 & 7/30 \\ 0 & 0 & 1 & 4/5 & 3/5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{matrix}$$

10. Extend the set $\{(1,0,1,0),\ (1,0,-1,0)\}$ to a basis of $\mathbb{R}^4$.

e_2, e_4

$u, v, w \to$ L.I

11. Let $u_1, v, w, x_1, x_2, x_3, x_4 \in \mathbb{R}^5$ satisfy $x_1 = u + 2v + 3w$, $x_2 = 2u - 3v + 4w$, $x_3 = -3u + 4v - 5w$ and $x_4 = -u + 6v + w$. Is $\{x_1, x_2, x_3, x_4\}$ linearly dependent?

$(\alpha+2\beta-3\delta-\delta)u+(2\alpha-3\beta+4\delta-6\delta)v+(3\alpha+4\beta-5\delta+\delta)w=0$

4 vectors in 3D space

$\Rightarrow$ linearly dependent

$$\begin{matrix} 1 & 2 & 3 & -1 \\ 2 & -3 & 4 & -6 \\ 3 & 4 & -5 & 1 \end{matrix} \to \begin{matrix} 1 & 0 & -11/17 & 3/17 \\ 0 & 1 & 2/17 & 4/17 \\ 0 & 0 & 47/17 & -18/17 \end{matrix}$$

12. Prove that the only proper subspaces of $\mathbb{R}^2$ are the straight lines passing through the origin.

$\dim = 0 \Rightarrow \{(0,0)\}$

$\dim = 1 \Rightarrow$ line through origin $\to$ proper subspace

$\dim = 2 \Rightarrow \mathbb{R}^2$ plane

13. Describe all subspaces of $\mathbb{R}^3$.

$\dim = 0 \Rightarrow$ origin

$\dim = 1 \Rightarrow$ lines through origin

$\dim = 2 \Rightarrow$ planes through origin

$\dim = 3 \Rightarrow$ full space ($\mathbb{R}^3$)

15. Let $A = \begin{bmatrix} 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ 1 & 1 & 1 & 2 & 1 \\ 5 & 8 & 5 & 7 & 5 \end{bmatrix}$. Write $r = \text{rank}(A)$ and $k = \text{null}(A)$.

(a) Find bases for $R(A)$ and $N(A)$, and then determine r and k.
(b) Express suitable $4 - r$ rows of A as linear combinations of other r rows.
(c) Express suitable $5 - r$ columns of A as linear combinations of other r columns.

$$\begin{matrix} 1 & 2 & 1 & 1 & 1 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & -20 & 2 & 0 & \end{matrix}$$

$$\begin{matrix} 1 & 0 & 1 & 3 & 1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{matrix}$$

R(A): (1,3,1,1,5), (2,5,1,1,8)

N(A): (1,0,-1,0,0), (3,-1,0,-1,0), (1,0,0,0,-1)

r=2, k=3

(c) $C_3 = C_1$, $C_4 = 3C_1 - C_2$, $C_5 = C_1$

(b) $R_3 = R_2 - 2R_1$, $R_5 = 2R_2 - R_1$

16. Construct bases for (i) the row space, (ii) the column space, and the null space for each of the following matrices:

(a) $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 7 \\ 2 & 4 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & 1 & -3 \\ 1 & 2 & 8 \\ 3 & -1 & 4 \\ 4 & -2 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 2 \\ 1 & 3 & -2 & 1 \end{bmatrix}$

(a) $\begin{matrix} 1 & 2 & 4 \\ 0 & -5 & -5 \\ 0 & 0 & 0 \end{matrix}$ → [124] [317]

$\begin{matrix} 1 & 2 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{matrix} \rightarrow \begin{matrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{matrix} \rightarrow [1\ 3\ 2]^t$ $[2\ 1\ 4]^t$

$x_1 + 2x_3 = 0$
$x_2 + x_3 = 0$

$[2,1,-1]^t$

$[-3\ 1\ 3\ 4]^t$
$[1\ 2\ -1\ -2]^t$
$[-3\ 8\ 4\ 2]^t$

(b) $\begin{matrix} 1 & 2 & 8 \\ -3 & 1 & -3 \\ 3 & -1 & 4 \\ 4 & -2 & 2 \end{matrix}$ $\begin{matrix} 1 & 2 & 8 \\ 0 & 7 & 21 \\ 0 & 0 & 1 \\ 0 & -10 & -30 \end{matrix}$ → $\begin{matrix} 1 & 2 & 8 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 3 \end{matrix} \rightarrow \begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{matrix}$

[-3,1,3], [1,2,8], [3,-1,4]

(c) $\begin{bmatrix}1&3&-2&1\\2&1&3&2\\3&4&5&6\end{bmatrix} \to \begin{bmatrix}1&3&-2&1\\0&-5&7&0\\0&-5&9&3\end{bmatrix} \to \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$ $\begin{matrix}43/10\\21/10\\15/10\end{matrix}$

row: $[3\ 4\ 5\ 6]$ $[2\ 1\ 3\ 2]$ $[1\ 3\ -2\ 1]$

column: $[3\ 2\ 1]^t$ $[4\ 1\ 3]^t$ $[5\ 3\ -2]^t$

null: $[4.3\ \ 2.1\ \ 1.5\ \ -1]^t$

17. Let $A \in \mathbb{F}^{n\times n}$. Consider A as the linear transformation $A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{n\times 1}$ given by $A(x) = Ax$ for each $x \in \mathbb{F}^{n\times 1}$. Prove the following:
    (a) A is one-one iff $N(A) = \{0\}$.
    (b) A is one-one and onto iff A maps any basis onto a basis of $\mathbb{F}^{n\times 1}$.
    (c) A is one-one iff A is onto.

(a) $Ax = Ay \Rightarrow x = y$

$Ax = 0 \Rightarrow x = 0 \Rightarrow N(A) = \{0\}$

(b) A is one-one $\Rightarrow$ columns of A are linearly independent → basis

A is onto $\Rightarrow$ column space spans $F^{n\times 1}$

(c) proved in B

18. Let A and B be matrices having the same RREF. Then which of the following may be concluded? Explain.
    (a) $R(A) = R(B)$ (b) $N(A) = N(B)$
    (c) The row space of A is equal to the row space of B.

(a) False (c) True

(b) True row operations don't change row spaces but can change column space

20. In the following a basis B for $\mathbb{F}^{3\times 1}$ and a vector $u \in \mathbb{F}^{3\times 1}$ are given. Compute the coordinate vector $[u]_B$.

(a) $B = \{[0,0,1]^t,\ [1,0,0]^t,\ [0,1,0]^t\},\ u = [1,2,3]^t$
(b) $B = \{[0,1,1]^t,\ [1,0,1]^t,\ [1,1,0]^t\},\ u = [2,1,3]^t$
(c) $B = \{[1,1,1]^t,\ [1,2,1]^t,\ [1,2,3]^t\},\ u = [3,2,1]^t$

(a) $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $\quad \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$

$\Rightarrow u_B = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$

(b) $\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \Rightarrow u_B = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$

(c) $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \Rightarrow u_B = \begin{bmatrix} -4 \\ -4 \\ 3 \end{bmatrix}$

14. Let $A \in \mathbb{F}^{n\times n}$. Let $\{v_1, \ldots, v_n\}$ be any basis of $\mathbb{F}^{n\times 1}$. Prove that A is invertible iff $Ax = v_i$ has at least one solution for each $i = 1, \ldots, n$.

If A is invertible; $x = A^{-1}v_i$

If $Ax = v_i$

let $v_i = e_i \Rightarrow Ax = e_i$

$\Rightarrow \operatorname{rank}(A|e_i) = \operatorname{rank}(A)$

$\downarrow \qquad \Downarrow$

$n \qquad n$

$\Rightarrow A$ is invertible

15. Let $A = \begin{bmatrix} 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ 1 & 1 & 1 & 2 & 1 \\ 5 & 8 & 5 & 7 & 5 \end{bmatrix}$. Write $r = \text{rank}(A)$ and $k = \text{null}(A)$.

(a) Find bases for $R(A)$ and $N(A)$, and then determine r and k.
(b) Express suitable $4 - r$ rows of A as linear combinations of other r rows.
(c) Express suitable $5 - r$ columns of A as linear combinations of other r columns.

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 1 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & -2 & 0 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 3 & 1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$r = 2 \quad k = 3$

$R(A) = [1\ 3\ 1\ 5]^t\ [2\ 5\ 1\ 8]^t$

$N(A) = [1\ 0\ -1\ 0\ 0]^t\ [3\ -1\ 0\ -1\ 0]^t$

$[1\ 0\ 0\ 0\ -1]^t$

$R_3 = R_2 - 2R_1 \quad R_4 = -R_1 + 2R_2$

$C_3 = C_1 \quad C_4 = 3C_1 - C_2 \quad C_5 = C_1$

17. Let $A \in \mathbb{F}^{n\times n}$. Consider A as the linear transformation $A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{n\times 1}$ given by $A(x) = Ax$ for each $x \in \mathbb{F}^{n\times 1}$. Prove the following:

(a) A is one-one iff $N(A) = \{0\}$.
(b) A is one-one and onto iff A maps any basis onto a basis of $\mathbb{F}^{n\times 1}$.
(c) A is one-one iff A is onto.

(a) $A(x_1) = Ax_1 \quad Ax_1 = Ax_2$

$A(x_2) = Ax_2 \quad \Rightarrow A(x_1 - x_2) = 0$

$Ax = 0 \Rightarrow x = 0$ (condition for injective)

$N(A) = \{x : A(x) = 0\} = \{0\}$

If $N(A)=\{0\} \Rightarrow$ The only solution of $Ax=0$ is $x=0$

$A(x_1-x_2)=0 \Rightarrow 0=x_1-x_2 \Rightarrow \boxed{x_1=x_2}$

$\Rightarrow$ injective

(b) Condition for surjection: Range $(A(x)=Ax)$ spans $F^{n\times 1}$.

If A maps one basis to another

→ Injection: no two distinct basis vectors map to the same vector

→ Surjection: Image spans F^n

If A is bijective:

$N(A)=\{0\}$, $\text{rank}(A)=n$

$\theta=\{\theta_1, \theta_2, \dots, \theta_n\}$ → basis

$\Rightarrow A\theta=\{A\theta_1, A\theta_2, \dots A\theta_n\}$

↓

linearly independent and spans F^n

(c) Rank Nullity theorm:

$N(A)=\{0\} \Rightarrow R(A)+N(A)=n \Rightarrow R(A)=n \Rightarrow$ Surjective

$R(A)=n \Rightarrow R(A)+n(A)=n \Rightarrow n(A)=0 \Rightarrow$ injective

11. Let $u_1, v, w, x_1, x_2, x_3, x_4 \in \mathbb{R}^5$ satisfy $x_1 = u + 2v + 3w$, $x_2 = 2u - 3v + 4w$, $x_3 = -3u + 4v - 5w$ and $x_4 = -u + 6v + w$. Is $\{x_1, x_2, x_3, x_4\}$ linearly dependent?

$x_1\ x_2\ x_3\ x_4$

let $T: V \to W$ $\quad u\ v\ w$ s.t $T(x_1) = u + 3v + 3w \cdots$

$$[T]_{CB} = [[Tx_1]_C \cdots [Tx_4]_C]$$

$$= \begin{bmatrix} 1 & 2 & -3 & -1 \\ 2 & -3 & 4 & 6 \\ 3 & 4 & -5 & 1 \end{bmatrix}$$

$\text{rank} \leq 3$

$\Rightarrow$ linearly dependent